

Mischa Lundberg

@mischalundberg,bsky.social • ORCID: 0000-0002-4816-785X

PROFESSIONAL SUMMARY

Highly skilled researcher and bioinformatician with extensive experience in Statistical Genetics and Machine Learning. Proven track record in managing complex genomic datasets, developing computational workflows for transposable element analysis, and leading multi-institutional research projects. Expertise spans from GWAS and Mendelian Randomisation to the development of scalable software tools for large-scale biological data analysis.

CAREER EXPERIENCE

Postdoctoral Researcher 11.2022 – Present

Institute of Biological Psychiatry, Roskilde, Denmark

- Led research on genetic architecture and biological psychiatry under A/Prof Dr. Andrew J. Schork.
- Developed and implemented methodologies for analysing large-scale biological data.
- Led the Copenhagen Hospital Biobank and Danish Blood Donor Study genetic ancestry assignment.
- Collaborated on high-impact research projects involving complex disease phenotypes.

Paternal leave: May – July 2025 (100%), Dec 2025 – May 2026 (100%)

PhD Student in Statistical Genetics (Bioinformatics)

10.2018 – 02.2023

Diamantina Institute – University of Queensland, Brisbane, Australia

- Dual Appointment: Conducted a shared PhD position between the University of Queensland and CSIRO (Health & Biosecurity).
- Conducted research on the causal relationships between bone mineral density, osteoarthritis, and chronic pain.
- Investigated novel Alzheimer's disease genes and epistasis using a machine-learning GWAS platform applied to the ADNI cohort.
- Executed a diverse range of projects utilising the UK Biobank (UKB) dataset, focusing on bone mineral density, osteoarthritis, and chronic pain.
- Supervised a PhD student during the final year of the program due to the primary supervisor's illness.
- Thesis on genetic association studies and multi-omics data integration.

Thesis Title: "Investigating a possible causal relationship between bone mineral density, osteoarthritis, and chronic pain"

Research Programmer 11.2015 – 12.2021

Mater Research Institute – University of Queensland, Brisbane, Australia

- Developed workflows and computational methods for transposable element (TE) analysis and simulation.
- Built automated pipelines for large-scale genomic data processing.
- Collaborated with senior researchers to implement novel bioinformatics algorithms.

CEO – LITUS Lundberg IT & Service 04.2009 – Present

Bensheim, Germany

- Manage a small technology firm with a focus on IT services.
- Oversee operations, business development, and team management.

LEADERSHIP & TEACHING

- **Guest Associate Editor:** Frontiers in Genetics (Research Topic: Genetic and Environmental Factors in Early-Life Development). 04.2025 – Present
- **Programming Tutor:** Taught C/C++ to Master's students at the Technical University of Darmstadt. 10.2010 – 04.2013
- **Committee Leadership:** Served as Co-Chair of the UQ Faculty of Medicine HDR student committee. 01.2020 – 03.2021

SKILLS

- Programming:
 - Python
 - R
 - Java
 - C++
 - C#
- Expertise:
 - GWAS
 - Polygenic Risk Scores
 - Mendelian Randomization
 - Statistical Genetics
 - Machine Learning
 - WES/Illumina/ONT alignment
 - Transposable element detection
 - Methylation calling
- Operating Systems:
 - Linux/Unix
 - MS Windows
 - MS Server

EDUCATION

PhD in Statistical Genetics
University of Queensland & The Commonwealth Scientific and Industrial Research Organisation
Australia
2018-2023

Bachelor of Arts in Informatics and Sociology
Technical University of Darmstadt
Germany
2009-2016

CERTIFICATIONS

Data protection officer - General Data Protection Regulation (GDPR)
07/2019
Datenschutz.com GmH, Germany
Expert in agile software engineering
11/2017
Andrena OBJECTS AG, Germany